



Roll Number

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2026
Even Semester - 2025-26
PH(CS)201 - ENGINEERING PHYSICS

TIME ALLOTTED: 3Hours

Maximum Marks : 70

Instructions to the candidate:*Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. In He-Ne LASER, neon atoms obtain energy (1) C01 BL1 PO1
(a) By collision with He atoms (b) From chemical reactions
(c) From optical pumping (d) From electrical pumping
2. In an OLED display, light is produced by (1) C04 BL2 PO1
(a) Liquid crystal molecules reflecting external light (b) Gas discharge inside plasma cells
(c) Organic materials emitting light when current passes through them (d) Electron beam striking a phosphor screen
3. In case of a p-n junction, if the reverse bias is increased, the width of the depletion region (1) C03 BL2 PO1
(a) Increases (b) Decreases
(c) Remains constant (d) None of these
4. The velocity with which energy is transported through a medium is (1) C05 BL1 PO1
(a) Group velocity (b) Phase velocity

- (c) Complex velocity (d) Imaginary velocity
5. Which of the following is not a characteristic of a "well-behaved" wave function? (1) C05 BL2 PO2
- (a) Single valued (b) Infinite in space
(c) Vanishes at boundary (d) Continuous first-order derivative
6. The working of a Cathode Ray Oscilloscope (CRO) is based on the deflection of (1) C03 BL1 PO1
- (a) X-rays (b) Electron beam
(c) Light beam (d) Ion beam
7. In an FCC lattice, the number of lattice points per unit cell is (1) C03 BL1 PO1
- (a) 1 (b) 2
(c) 3 (d) 4
8. Lissajous pattern obtained on a CRO can be used to determine (1) C04 BL2 PO2
- (a) Phase Shift (b) Voltage Amplitude
(c) Amplitude Distortion (d) None of these
9. For a non-dispersive medium (1) C05 BL1 PO1
- (a) $V_g = V_p$ (b) $V_g < V_p$
(c) $V_g > V_p$ (d) none of these
10. The number of Bravais lattices in three dimensions are (1) C03 BL1 PO1
- (a) 7 (b) 9
(c) 14 (d) 19
11. Size confinement is maximum in (1) C02 BL2 PO1
- (a) Bulk material (b) Quantum dot
(c) Quantum wire (d) Quantum well
12. A quantum wire is an example of a: (1) C01 BL1 PO1
- (a) 0D nanomaterial (b) 1D nanomaterial
(c) 2D nanomaterial (d) 3D bulk material

Part B**(Answer any three of the following) 3x5=15**

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|---|-----|-----|-----|-----|
| 13. What do you mean by group and phase velocity? Derive the relation between them. Mention how it gets modified for a non-dispersive medium. | (5) | C05 | BL4 | PO2 |
| 14. Explain the working of a He-Ne laser with necessary energy level diagram. | (5) | C01 | BL6 | PO1 |
| 15. Write down the advantages and disadvantages of magnetic storage devices. | (5) | C04 | BL2 | PO1 |
| 16. (a) State Bragg's law of Diffraction. | (2) | C03 | BL1 | PO1 |
| (b) The spacing between principle planes of NaCl crystal is 2.82 Å. It is found that first order order Bragg reflection occurs at an angle of 100°. What is the wavelength of incident ray? | (3) | C03 | BL3 | PO2 |
| 17. Write down the working principle of CRO? | (5) | C04 | BL2 | PO1 |

Part C**(Answer any three of the following) 3x15=45**

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| 18. (a) Derive Einstein A and B coefficient relations. State the physical significance of Einstein A and B coefficients | (10) | C01 | BL5 | PO1 |
| (b) Explain the working principle of a semiconductor laser | (5) | C01 | BL2 | PO2 |
| 19. (a) Normalize the wave function $\psi = A \sin(\pi x/l)$ for the region $[-l, +l]$ | (5) | C05 | BL4 | PO2 |
| (b) Explain from the Uncertainty Principle why an electron cannot exist inside the nucleus. | (5) | C05 | BL3 | PO2 |
| (c) Write short notes on any one:
CNT
Nanomaterials
Quantum Wire
Quantum Dot. | (5) | C02 | BL3 | PO1 |
| 20. (a) Distinguish between crystalline and amorphous Solids. | (5) | C03 | BL4 | PO2 |
| (b) Deduce the inter-planner spacing of a simple cubic lattice of side a | (5) | C03 | BL4 | PO2 |

(c) Prove that in a cubic crystal, the direction $[hkl]$ is perpendicular to the plane (hkl) .	(5)	C03	BL5	PO2
21. (a) Explain the Uncertainty Principle	(5)	C05	BL3	PO2
(b) Show that Uncertainty Principle is valid for a moving electron but not for moving car	(5)	C05	BL3	PO2
(c) Explain the physical interpretation of the wave function	(5)	C05	BL4	PO2
22. (a) Explain the concept of quantum size effect in nanomaterials.	(5)	C01	BL2	PO1
(b) Describe the structure and potential applications of Carbon Nanotubes (CNTs).	(5)	C01	BL1	PO1
(c) Analyze the operation principle of a Plasma display panel.	(5)	C03	BL4	PO1